A380X ELEC, fuel and FCOM gap alerts

The A380X ECAM raises the FCOM combined busbar alerts, ELEC STATIC INV FAULT and the FUEL wing feed pump, wings balance and inner tank alerts, on top of the ELEC alerts and the A380X failures merged together; since it also holds the A380X part (new failures, ECAM alerts, memos, voices and system models: air data, cabin pressure, drive oil, APU, smoke, bleed, gear and brakes, engines, fuel, anti-ice, oxygen, avionics network, GPWS and windshear), the XFR pushbuttons and other fixes

Aircraft	A380X (A32NX: ADIRU power supplies, engine anti-ice valves and anti-ice key events, through shared code)
Where to find it	flyPad > Failures (ATA 21, 24, 26, 27, 28, 29, 30, 31, 32, 34, 35, 36, 42, 49, 73, 79); overhead ELEC, FUEL (XFR, EMER OUTR TK XFR), AIR (ENG BLEED), ANTI ICE, CARGO SMOKE, APU, EVAC and ADIRS panels; E/WD, STATUS (and STATUS MORE), PFD, SD ELEC, FUEL, ENGINE, PRESS, WHEEL and DOOR/OXY pages; ECP; T.O CONFIG key; flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs (with core/efb-redesign)
Branch	`core/a380x-elec-fuel-alerts`, based on `core/a380x-elec` with `core/failures-a380x` merged in (stacked: needs both merged first; core/a380x-elec itself needs core/surveillance, core/a380x-mfd-fms and core/gps). Why a separate branch: the combined busbar and wing feed alerts need the ELEC alerts of core/a380x-elec and the FWS of core/failures-a380x, and merging either of those two into the other would put each scope into the other. The merge keeps both failure lists, both flyPad realism settings and the ECAM texts of both once, and applies the merge notes of core/failures-a380x: T.O CONFIG NORMAL reads the ELEC GEN / DRIVE alerts, ELEC EMER CONFIG reads the FWS signal inhibited by ALL ENG FLAME OUT; the FUEL JETTISON pbs are still read directly (no core/fuel here). Since `core/fcom-audit-fixes` is merged in too (merge f0aaa86dc), a third parent: the A380X door, A-ICE and NAV alert files of that branch are the ones its later alerts extend. This branch is the A380X FCOM gap alert branch: it is the only one holding all the A380X alert code the later alerts build on (ELEC, XPDR and SURV of core/a380x-elec, engine, brake, air data and fire of core/failures-a380x, ECAM of core/ecam-fws). Merge core/a380x-elec, core/failures-a380x and core/fcom-audit-fixes into it again when they change. Dependencies on other core branches (merge both): the A380X smoke detection model and the CARGO SMOKE panel lights are on `core/failures-a32nx` (the SMOKE alerts here read its L:vars); the flyPad pin option of the smart FIVE HUNDRED callout is on `core/efb-redesign`. Merge note with `core/fuel`: the fuel transfer memos and the manual transfers of this branch read L:A380X_FUEL_JETTISON_IN_PROGRESS directly; with core/fuel they take its jettison signals. The features of the parents are described in their own manuals; the A32NX parts of these alerts are on core/failures-a32nx.
Status	Tested in the sim by the user on a combined test build: ELEC AC BUS 2+4 FAULT OK (flight) and FUEL L WING FEED PMPs FAULT OK. The other combined busbar alerts, STATIC INV FAULT, WINGS NOT BALANCED / BALANCED and INR TKs QTY LO: not sim-tested (unit tests FwsElecCombinedBusAlerts.spec.ts, FwsFuelFeedBalanceAlerts.spec.ts). On a combined test build: T.O CONFIG NORMAL with GEN 3 OFF recorded as passed; no ELEC EMER CONFIG during ALL ENG FLAME OUT tested. The core branch itself was not sim-tested. A full MSFS restart is needed after installing. The FCOM alert additions: not sim-tested; unit/Rust tests only.
Pictures	No pictures: the alerts use the stock E/WD, STATUS and SD pages.

1. Combined ELEC busbar alerts and ELEC STATIC INV FAULT
2. FUEL wing feed pumps, wings balance and inner tank alerts
3. ELEC alerts with the A380X failures: T.O CONFIG and ELEC EMER CONFIG
4. ADR and IR failures, ADIRU power supplies
5. CAB PRESS, RAM AIR, APU FAULT, HYD, A-ICE and SURV alerts
6. PFD FMA messages, AVAIL, EGT start limits and SD items
7. smart FIVE HUNDRED callout
8. ECAM advisories and STATUS MORE
9. EVAC ACTIVE memo
10. cabin pressure, generator drive oil, ELEC SD failures, e-pump inhibition
11. APU automatic shutdowns, smoke detection and cargo extinguishing alerts
12. bleed monitoring, leak detection, BLEED FAULT and LEAK alerts

13. landing gear and brakes
14. HP fuel valve, FADEC faults, motoring and auto-idle
15. LP valve, emergency outer tank transfer and fuel leak detection
16. anti-ice valves, wing anti-ice, oxygen, anti-ice key events
17. avionics network failures, SFCC channels, TAIL STRIKE
18. reactive windshear, GPWS mode 2, alpha floor Mach limit
19. NAV FM / GPS POS DISAGREE and NAV LS TUNING DISAGREE
20. A380X FWS voices: sidestick priority, intermediate RA callouts, WINDSHEAR, TERRAIN, GLIDE SLOPE, PITCH PITCH, COMPANY MESSAGE
21. No BRAKES BTV FAULT while an inner engine is not running
22. Engine exceedance marks only on displayed values
23. XFR pushbuttons start in AUTO and drive the manual fuel transfers

1. Combined ELEC busbar alerts and ELEC STATIC INV FAULT

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24; E/WD, STATUS, SD ELEC AC / DC | **Status:** ELEC AC BUS 2+4 FAULT tested in the sim by the user on a combined test build: OK (flight). The other combined alerts and STATIC INV FAULT: not sim-tested (unit tests FwsElecCombinedBusAlerts.spec.ts).

What it is for

A380 FCOM PRO-ABN-ECAM-10-24 has combined alerts for the loss of several busbars together. The A380X showed the single busbar alerts side by side instead. Six combined alerts now trigger when all their busbars are lost and replace the single alerts they cover, with their procedure lines, flight phase inhibitions (phases 4, 5 and 10) and STATUS:

Alert	Busbars lost
ELEC AC BUS 1+2 & DC BUS 1 FAULT	AC 1, AC 2, DC 1
ELEC AC BUS 2+3 & DC BUS 1+2 FAULT	AC 2, AC 3, DC 1, DC 2
ELEC AC BUS 2+4 FAULT	AC 2, AC 4
ELEC AC BUS 3+4 & DC BUS 2 FAULT	AC 3, AC 4, DC 2
ELEC DC BUS 1+2 FAULT	DC 1, DC 2
ELEC DC BUS 1+ESS FAULT	DC 1, DC ESS

- A combined alert whose busbars are all part of another active one is replaced by it (DC BUS 1+2 FAULT inside AC BUS 2+3 & DC BUS 1+2 FAULT); combined alerts that only overlap are both shown.
- The fuel lines of these procedures depend on the flight time above FL 300 since departure (more or less than 30 min, counted from lift-off) and lead to gravity feeding (descent, CROSSFEED ... OFF, EXPECT NORM COLLECTOR CELL DEPLETION, AVOID NEG G LOAD), shown in flight only.
- The STATUS lists the FCOM lines that have an ECAM text, for example AC BUS 2+4 FAULT: MAX SPEED 310 KT, FUEL PARTLY UNUSABLE; INOP SYS F/CTL PROT, AP 1+2, A/THR, FD 1+2, ADR 2+3, TCAS 1+2, SATCOM, JETTISON, FEED TK 1 and 4 PMPs; INFO ALTN LAW : PROT LOST, FEED TK 1 (4) BY GRVTY ONLY. The JETTISON INOP SYS text (also in AC BUS 2+4, DC BUS 1+2 and DC BUS 1+ESS FAULT) is added here, identical to the one of `core/fuel`.
- **ELEC STATIC INV FAULT** (crew awareness, phases 3 to 10 inhibited): the static inverter runs only on batteries (no AC ESS power, both battery contactors closed); it counts as failed when it runs but gives no normal output.

Limitations and why

Limitation	Why	What to do
Some FCOM STATUS lines are missing (for example ATC COM VOICE ONLY, VHF 3, MOST FUEL XFR PMPs).	They have no ECAM text in FBW yet.	-

REDUND LOSS lines are not shown for these alerts.	They never reach the STATUS page in the FBW FWS.	-
The combined alerts do not trigger in ELEC EMER CONFIG.	Design choice: the bus fault alerts need one AC busbar powered; ELEC EMER CONFIG covers that case.	-
VENT AVNCS EXTRACT OVRD and OUTR TK XFR MAN lines are never ticked.	No avionics ventilation pb-sw; the manual outer tank transfer pb-sw is inoperative in the sim.	-

How to use

1. When a combined ELEC alert shows, follow its E/WD procedure (the sensed lines tick as you act).
2. Read the STATUS page; in flight follow the gravity feeding lines if they show.

How to test

Setup: cruise above FL300, AP ON.

#	Do this	Expected result
1	flyPad > Failures > ATA 24 Electrical Power > **AC 2**, then **AC 4**.	Master caution and single chime; E/WD ELEC AC BUS 2+4 FAULT, not the single AC BUS 2 / AC BUS 4 alerts; SD ELEC AC page.
2	ECAM control panel: STS.	STATUS: ALTN LAW : PROT LOST, MAX SPEED 310 KT, INOP SYS among others JETTISON.
3	Deactivate both. Fail **DC 1**, then **DC 2**.	ELEC DC BUS 1+2 FAULT instead of DC BUS 1 / DC BUS 2 FAULT. Deactivate both.
4	On the ground, aircraft on batteries only (BAT pbs AUTO, no EXT PWR, no APU). Fail **Static Inverter**.	ELEC STATIC INV FAULT. Deactivate it.

2. FUEL wing feed pumps, wings balance and inner tank alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; overhead FUEL panel; E/WD, SD FUEL | **Status:** FUEL L WING FEED PMPs FAULT tested in the sim by the user on a combined test build: OK. WINGS NOT BALANCED / BALANCED and INR TKs QTY LO: not sim-tested (unit tests FwsFuelFeedBalanceAlerts.spec.ts).

What it is for

Four FUEL alerts of A380 FCOM PRO-ABN-ECAM-10-28 with their crossfeed procedure lines:

- **FUEL L (R) WING FEED PMPs FAULT**: all pumps of feed tanks 1 and 2 (left) or 3 and 4 (right) at low pressure, abnormally not running or switched off. It replaces the two FEED TK n MAIN + STBY PMPs FAULT alerts of its tanks. Inhibited in phases 4 to 7, 9 and 10. Pumps switched off on request of FUEL WINGS NOT BALANCED do not count.
- **FUEL WINGS NOT BALANCED**: on the ground, an imbalance above 6 600 lb between the symmetric feed tanks or the inner tanks, 5 500 lb between the mid tanks or 2 600 lb between the outer tanks (not while refuelling); in flight, more than 6 600 lb between feed tanks 1 and 4 or 2 and 3. In flight it stays until that pair is balanced (the lighter tank is no longer lighter), then **FUEL WINGS BALANCED** shows until its pumps are back ON and its crossfeeds OFF. Inhibited in phases 3 to 7, 9 and 10.
- **FUEL INR TKs QTY LO**: fuel on board above 110 200 lb with an inner tank below 11 200 lb (ZFW and ZFCG entered, both outer tanks full). Inhibited in phases 3 to 11.

Limitations and why

Limitation	Why	What to do
The "balanced" point of WINGS NOT BALANCED is a design choice.	The FCOM gives no balanced threshold.	-
Landing ends the in-flight balance procedures.	Design choice.	-

How to use

1. When a FUEL alert shows, follow its E/WD lines with the CROSSFEED and feed pump pbs of the overhead FUEL panel.
2. Watch the feed tank quantities on the SD FUEL page until the alert clears.

How to test

Setup: cruise.

#	Do this	Expected result
1	flyPad > Failures > ATA 28 Fuel > **Feed tank 1 main pump**, **Feed tank 1 standby pump**, **Feed tank 2 main pump** and **Feed tank 2 standby pump**.	FUEL L WING FEED PMPs FAULT replaces the two FEED TK MAIN + STBY PMPs alerts.
2	Follow the E/WD lines (CROSSFEED pbs on the overhead FUEL panel).	The crossfeed lines tick.
3	Deactivate the four failures; CROSSFEED pbs back to normal.	The alert goes.
4	On the ground, engines running, set the left outer tank about 1.5 t (3 300 lb) above the right one (fuel set per tank, for example with the MSFS fuel menu).	FUEL WINGS NOT BALANCED. Set the tanks equal: it goes.

3. ELEC alerts with the A380X failures: T.O CONFIG and ELEC EMER CONFIG

> **Aircraft:** A380X | **Where:** T.O CONFIG key; E/WD | **Status:** On a combined test build: T.O CONFIG NORMAL blocked by GEN 3 OFF recorded as passed in the commit message; no ELEC EMER CONFIG during ALL ENG FLAME OUT tested OK. The core branch itself was not sim-tested.

What it is for

What this branch adds by having both parents together:

- **T.O CONFIG NORMAL** (the A380 rule of `core/failures-a380x`) also checks the ELEC GEN 1 (2)(3)(4) FAULT, GEN OFF and DRIVE DISCONNECTED alerts of `core/a380x-elec`, as in the FCOM list (DSC-31-40-10).
- **ELEC EMER CONFIG** (of `core/a380x-elec`) is not shown while ENG ALL ENG FLAME OUT is (FCOM: "This alert inhibits the ELEC EMER CONFIG alert").

How to use

1. Before take-off press T.O CONFIG with all four generators ON.
2. In an all engines flame out, follow ALL ENG FLAME OUT; ELEC EMER CONFIG does not come on top of it.

How to test

#	Do this	Expected result

1	On the ground, four engines running, ready for take-off. Overhead ELEC panel: GEN 3 pb OFF. ECAM control panel: T.O CONFIG key.	ELEC GEN 3 OFF; no T.O CONFIG NORMAL.
2	GEN 3 pb ON, wait 5 s, T.O CONFIG key again.	T.O CONFIG NORMAL.
3	In flight at FL200, 300 kt, APU OFF: activate the four "Engine n flameout (crew relight possible)" failures (flyPad > Failures > ATA 72). Wait 20 s.	Red ENG ALL ENG FLAME OUT; the RAT deploys; no ELEC EMER CONFIG. Relight as in the core/failures-a380x manual.

4. ADR and IR failures, ADIRU power supplies

> **Aircraft:** A380X and A32NX (ADIRU power supplies) | **Where:** flyPad > Failures > ATA 34; overhead ADIRS panel; E/WD | **Status:** Not sim-tested (unit tests: adirs.rs 17 Rust tests, FwsNavAlerts.spec.ts, failure id specs).

What it is for

Air data and inertial data could not be lost on their own, and the ADIRUs ran without their busbars. Now:

- Six new flyPad failures: **ADR 1 (2)(3) (air data lost, FAULT light)** and **IR 1 (2)(3) (inertial data lost, FAULT light)**. The outputs of that ADIRU part go to failure warning and its FAULT light comes on (new L:A32NX_ADIRS_ADR_n_FAULT_WARN_DISCRETE, shared ADIRS model).
- Each ADIRU is supplied by its busbars, from the FCOM ELEC INOP SYS lists: the ADR needs its AC and its DC busbar, the IR runs on either. Power cuts up to 200 ms, and the first 5 s after a flight load, are ignored, so a flight loaded on the runway or in the air keeps its alignment.
- An ADR that lost its DC busbar no longer gives a NAV ADR FAULT on top of the ELEC busbar alert.
- Shared code: the A32NX ADIRU supplies come with it (the ADIRS model now needs each aircraft's supplies). The A32NX ADR / IR failures, ON BAT light and alerts are on `core/failures-a32nx`.

What you should see

Situation	What you should see
ADR 1 failed	ADR 1 FAULT light on the ADIRS panel; E/WD NAV ADR 1 FAULT; the air data of ADR 1 are lost.
IR 1 failed	IR 1 FAULT light; E/WD NAV IR 1 FAULT; the inertial data of IR 1 are lost.
An ADR busbar lost (ELEC alert)	That ADR is lost, but only the ELEC alert shows, no NAV ADR FAULT.

Limitations and why

Limitation	Why	What to do
No high angle of attack / sideslip thresholds for the A380X ADRs.	The FCOM gives no boundary.	-
The A32NX IR 10 min alignment is not on this branch.	It is on core/failures-a32nx only (merge note: keep it when both are merged).	-

How to use

1. Open flyPad > Failures > ATA 34 Navigation > and activate an ADR or IR failure.
2. Follow the NAV procedure on the E/WD (ADIRS panel and source switching).

How to test

#	Do this	Expected result
1	A380X in cruise, AP ON. Activate **ADR 1 (air data lost, FAULT light)**.	ADR 1 FAULT light; NAV ADR 1 FAULT on the E/WD. Deactivate it.
2	Activate **IR 2 (inertial data lost, FAULT light)**.	IR 2 FAULT light; NAV IR 2 FAULT. Deactivate it.
3	Load a flight on the runway, then another one in cruise.	The ADIRUs stay aligned (no ALIGN, attitude and navigation available at once).

What PR testers should check

- Flights loaded on the runway and in cruise keep the IR alignment.
- The ADR and IR failures from the flyPad (lights, alerts, lost data).

5. CAB PRESS, RAM AIR, APU FAULT, HYD, A-ICE and SURV alerts

> **Aircraft:** A380X | **Where:** E/WD, STATUS, memos; SD PRESS; overhead RAM AIR, APU, ANTI ICE and OXYGEN panels | **Status:** Not sim-tested (unit tests: 8 new spec files CabPressAlerts, AutoFlightAlerts, ApuFaultAlerts, FuelTransferMemos, HydPumpStatus, AntiIceAlerts, CabinOxygenMasks, AbnormBleedConfig, and 5 tests in FwsSurvAlerts.spec.ts, all failing first).

What it is for

A380X FCOM alerts and memos that were missing or never triggered:

- **CAB PRESS** (FCOM PRO-ABN-ECAM-10-21-30): EXCESS NEGATIVE DIFF PRESS (red warning, phases 8 and 9) is held until the differential pressure is back above -0.32 PSI; DIFF PRESS HI (phase 8) only in automatic mode; DIFF PRESS LO in phases 8 and 9; new **CAB PRESS LDG ELEVN FAULT** (phase 8) when neither FMS sends a landing elevation (no destination), with the LDG ELEVN deferred procedure.
- **RAM AIR** memo with the RAM AIR pb ON (DSC-21-10-20): green on the ground with no engine running, amber otherwise.
- **AUTO FLT APPROACH CAPABILITY DOWNGRADED** (PRO-ABN-ECAM-10-22) when RA SYS B alone is failed: STATUS CAT 3 DUAL inoperative, CAT 3 SINGLE ONLY.
- **APU FAULT** (PRO-ABN-ECAM-10-49) after an APU emergency or automatic shutdown by its ECB: EMER SHUTDOWN or AUTO SHUTDOWN, -APU MASTER SW ... OFF; STATUS INOP SYS APU. Not shown while APU FIRE is.
- **Fuel memos** (DSC-28-20): FWD XFR IN PROGRESS (fuel leaving the trim tank), OUTR TK XFR IN PROG (fuel leaving the outer tanks), OUTR TKS XFRD and TRIM TK XFRD (tanks emptied). The fuel model tells no transfer state, so the FWS watches the tank weights: more than 5 kg out of a tank in 10 s is a transfer.
- **HYD:** ENG PMP PRESS LO lists the pumps of its system under STATUS INOP SYS (two or more failed) or REDUND LOSS (one failed); G (Y) RSVR AIR PRESS LO and G (Y) SYS OVHT get their FOR APPROACH deferred procedures (engine pumps back ON); G+Y SYS PRESS LO shows LDG PERF AFFECTED and the FCOM INOP SYS list (BTV, CAT 2, GLS AUTOLAND), without STABILIZER.
- **A-ICE ICE DETECTED** and **A-ICE SEVERE ICE DETECTED** (PRO-ABN-ECAM-10-30), cruise only (phase 8): ENG 1-4 A-ICE ... ON, and WING A-ICE ... ON for severe icing.
- **Cabin oxygen masks** drop by themselves above 13 800 ft cabin altitude (DSC-35-30-10, supplied by DC ESS).
- **SURV GPWS 1 (2)(1+2) FAULT** phases and lines; **SURV TAWS 1 (2)(1+2) FAULT** now triggers when the GPWS and the TERR SYS of a system are both failed, instead of the two separate alerts.
- **AIR ABNORM BLEED CONFIG** after ENG SHUT DOWN or ENG FIRE with an engine bleed lost: all the FCOM lines (XBLEED, AIR FLOW, PACK, WING A-ICE). The title is shown (the FWS has no procedure without a title).

Limitations and why

Limitation	Why	What to do
The fuel transfer memos come from the tank weights (5 kg in 10 s).	The fuel model gives no transfer state; the FCOM gives no detection rule (design choice).	-
No COLDFUEL OUTR TK XFR or GND XFR IN PROGRESS memo.	No fuel temperature model, no ground transfer function.	-
APU FAULT: the "failure but APU still available" and "failed start above FL 150" lines stay hidden.	The ECB tells neither case apart.	-
APPROACH CAPABILITY DOWNGRADED triggers only on RA SYS B.	The other FCOM triggers are PRIM monitorings the sim does not have.	-
The fuel transfer memos read L:A380X_FUEL_JETTISON_IN_PROGRESS directly.	The jettison signals of core/fuel are not on this branch.	With core/fuel merged they use its signals.

How to use

1. Nothing to set: the alerts and memos come by themselves; follow the E/WD lines and read the STATUS page.
2. For a FOR APPROACH deferred procedure, apply it in approach as the ECAM shows.

How to test

#	Do this	Expected result
1	A380X at the gate, engines off. Overhead AIR panel: RAM AIR pb ON.	Green RAM AIR memo. RAM AIR OFF.
2	Cruise, AP ON. flyPad > Failures > ATA 34 Navigation > **RA SYS B**.	AUTO FLT APPROACH CAPABILITY DOWNGRADED; STATUS: CAT 3 DUAL inoperative, CAT 3 SINGLE ONLY. Deactivate it.
3	Cruise. flyPad > Failures > ATA 34 Navigation > **GPWS 1**.	SURV GPWS 1 FAULT.
4	Also activate **TERR SYS 1**.	SURV TAWS 1 FAULT instead of SURV GPWS 1 FAULT and SURV TERR SYS 1 FAULT. Deactivate both.
5	Cruise. flyPad > Failures > ATA 29 Hydraulic Power > **Green reservoir air leak**; wait for HYD G RSVR AIR PRESS LO and follow it.	The FOR APPROACH deferred procedure G RSVR AIR PRESS LO (G ENG 1 PMP A AND B ... ON, G ENG 2 PMP A AND B ... ON) is listed.
6	Cruise above FL 300. flyPad > Failures > ATA 21 Air Conditioning > **Rapid Decompression**.	When the cabin altitude passes 13 800 ft the cabin oxygen masks drop by themselves; CAB PRESS alerts as before. Deactivate it, descend.
7	Long cruise with fuel in the outer tanks.	OUTR TK XFR IN PROG while the outer tanks empty, then OUTR TKS XFRD.
8	Cruise, no destination in the FMS (for example a flight plan without destination).	CAB PRESS LDG ELEVN FAULT (phase 8 only).

What PR testers should check

- The fuel transfer detection (5 kg in 10 s): no false memo in normal burn, memos during real transfers.
- The masks drop above 13 800 ft cabin altitude, not below.
- GPWS / TERR SYS / TAWS combinations of the SURV alerts.

6. PFD FMA messages, AVAIL, EGT start limits and SD items

> **Aircraft:** A380X | **Where:** PFD FMA; E/WD engine gauges; SD FUEL and WHEEL pages; E/WD | **Status:** Not sim-tested (unit tests: 11 spec files, 66 tests, all failing first, among them AvailIndication, EgtLimits, FMADefinitions, FuelImbalanceIndication, LgersIndications, ata70).

What it is for

- **PFD FMA** line 3: CHECK SPD MODE (its L:var was already set by the FMC) and CHECK EO SPD SETTING now show.
- **E/WD AVAIL**: stays 10 s after an engine start on the ground, pulses 1 min after a relight in flight, goes when the thrust lever is set above idle.
- **E/WD EGT**: the amber limit is 745 C during a ground start and 865 C during a relight in flight; no EEC trim during a start.
- **SD FUEL**: the tank quantities of an unbalanced pair are underlined amber, with the trigger of FUEL WINGS NOT BALANCED.
- **SD WHEEL**: L/G CTL and L/G GRVTY EXTN IN PROGRESS now show.
- **ENG 2 (3) SHUT DOWN**: the landing distance line reads LDG DIST IMPACT ON WET/CONTAM RWY ONLY.

Limitations and why

Limitation	Why	What to do
No EXTEND / RETRACT SPD BRK or CHECK APPR SEL on the A380X.	They need VNAV outputs the FMS does not give.	-
No WHEEL UP LOCK indication.	The gear model has no uplock state.	-
The ENG SHUT DOWN bleed lines are not added.	Not done yet.	-

How to use

1. Nothing to set: the indications follow the FCOM by themselves.

How to test

#	Do this	Expected result
1	A380X on the ground, engine start (ENG START selector, ENG MASTER 1 ON).	EGT amber limit at 745 C during the start; AVAIL shows when the engine is started and goes 10 s later.
2	On the ground, make one feed tank more than 6 600 lb heavier than its symmetric tank (MSFS fuel menu). SD FUEL page.	The quantities of that pair are underlined amber, with FUEL WINGS NOT BALANCED.
3	In flight, gear up: L/G GRVTY switch DOWN. SD WHEEL page.	L/G GRVTY EXTN IN PROGRESS until the gear is down.
4	Cruise: ENG MASTER 2 OFF (shut down engine 2).	ENG 2 SHUT DOWN; its landing distance line reads LDG DIST IMPACT ON WET/CONTAM RWY ONLY. Relight.

What PR testers should check

- The width of the fuel underline on the SD FUEL page.
- The EGT colour during a hot start.
- The AVAIL and fuel pulse animations.

7. smart FIVE HUNDRED callout

> **Aircraft:** A380X | **Where:** Aural callouts; flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs | **Status:** Not sim-tested (unit tests: LegacyGpws.spec.ts smart 500 block, failing without the change).

What it is for

The A380 FCOM (DSC-34-10-50 AUTOMATIC CALLOUTS) has a pin programmed "smart" FIVE HUNDRED callout: it is called only when the aircraft is more than half a dot off the glide slope, or without a valid glide slope. The callout logic (AutoCallOuts isFiveHundredCalloutDue, read by LegacyGpws) is on this branch.

Limitations and why

Limitation	Why	What to do
The flyPad option that sets the pin is not on this branch.	The flyPad settings pages are on core/efb-redesign.	Merge both: core/efb-redesign has the "Inhibited When Glideslope Captured" option.

How to use

1. flyPad > Settings > Aircraft Options / Pin Programs > Automatic Call Outs: under FIVE HUNDRED, set "Inhibited When Glideslope Captured" (needs core/efb-redesign).
2. Fly the approach: FIVE HUNDRED is only called when off the glide slope.

How to test

#	Do this	Expected result
1	With core/efb-redesign merged: set the option. ILS approach on the glide slope.	No FIVE HUNDRED callout at 500 ft RA.
2	Second approach, more than half a dot above or below the glide slope at 500 ft (or a non-ILS approach).	FIVE HUNDRED is called.
3	Option off, FIVE HUNDRED callout on.	FIVE HUNDRED is called on every approach.

What PR testers should check

- The callout on and off the glide slope with the pin option set and not set.

8. ECAM advisories and STATUS MORE

> **Aircraft:** A380X | **Where:** SD ENGINE and PRESS pages; ECP ENG and PRESS pbs, MORE pb; E/WD ADV; STATUS MORE | **Status:** Not sim-tested (unit tests: FwsSystemDisplayLogic.spec.ts and StatusMore.spec.ts, failing first, and FwsAdvisories.spec.ts).

What it is for

- **ECAM advisories** (FCOM DSC-31-40-10 ADVISORY): when an ENGINE value (oil quantity, oil temperature above 163 C, a rotor vibration above 5 units) or the CAB PRESS cabin V/S (above 1 800 ft/min) pulses on its SD page, the SD calls that page if it shows a flight phase page and its ECP pb comes on; if the crew had selected another page, the ECP pb flashes. The E/WD shows a pulsing ADV with the memos. ADV goes when the ECP pb is pressed or the value is back in range. New L:A380X_ECAM_SD_ADVISORY_PB.
- **STATUS MORE** (ECP MORE pb, DSC-31-40-10): new ALERTS IMPACTING LDG PERF section (the active alerts with a landing performance line), and the MORE INFO line of MISC OIS FAULT.

Limitations and why

Limitation	Why	What to do
No advisory for the nacelle temperature, delta P, cabin altitude or APU OIL LVL LO.	The SD pages do not pulse these values yet.	-

How to use

1. When ADV pulses on the E/WD, look at the called SD page; press its ECP pb to acknowledge.
2. When MORE shows at the bottom of the STATUS page, press the ECP MORE pb.

How to test

#	Do this	Expected result
1	A380X on the ground, engines at idle, SD on the flight phase page. flyPad > Failures > ATA 79 Oil > **Engine 1 oil overheat**; thrust lever 1 forward.	When the oil temperature passes 163 C and pulses, the SD shows the ENGINE page, the ECP ENG pb comes on, ADV pulses on the E/WD (with memos).
2	Press the ECP ENG pb.	ADV goes. Lever idle, deactivate the failure.
3	Flight with an alert that has a landing performance line (for example ENG 2 SHUT DOWN). STATUS, then ECP MORE pb.	STATUS MORE lists the alert under ALERTS IMPACTING LDG PERF.

What PR testers should check

- No ENG advisory at flight load while the oil quantity still reads 0.

9. EVAC ACTIVE memo

> **Aircraft:** A380X | **Where:** Overhead EVAC panel; E/WD memo | **Status:** Not sim-tested (unit tests: FwsMemosEvac.spec.ts).

What it is for

The A380 FCOM lists an EVAC ACTIVE memo while the evacuation command is on. The E/WD now shows EVAC ACTIVE while the EVAC COMMAND pb-sw is ON.

Limitations and why

Limitation	Why	What to do
The memo colour is green.	Assumed: the FCOM does not give it.	-

How to use

1. Set the EVAC COMMAND pb-sw ON (overhead EVAC panel): the memo shows until it is set OFF.

How to test

#	Do this	Expected result
1	A380X at the gate, powered. EVAC COMMAND pb-sw ON.	EVAC ACTIVE memo on the E/WD.
2	EVAC COMMAND pb-sw OFF.	The memo goes.

What PR testers should check

- The EVAC ACTIVE memo with the EVAC COMMAND pb-sw.

10. cabin pressure, generator drive oil, ELEC SD failures, e-pump inhibition

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 24; overhead PRESS, ELEC (DRIVE, APU GEN) and HYD panels; SD ELEC AC / DC; E/WD | **Status:** Not sim-tested (unit tests: CPC, DirectDrive, A380 pressurisation, fault light and e-pump Rust tests, failing first; FwsElecDriveOilAlerts.spec.ts and ElecFailureStates.spec.ts).

What it is for

- **Manual cabin pressure**: in MAN, the cabin altitude target holds the current cabin altitude and the V/S target 0 ft/min until the CABIN ALT TRGT or V/S TRGT selector is turned (it jumped to the selector position). Without the FMS landing elevation the controllers regulate to 7000 ft in descent (7500 ft also without ADIRU pressure) instead of 0 ft, and tell the FWS (new CPCS word bit 19).
- **Generator drive oil** (FCOM PRO-ABN-ECAM-10-24): new flyPad failures **DRIVE 1 (2)(3)(4) Oil Leak** and **DRIVE 1 (2)(3)(4) Oil Cooling Lost**; ELEC DRIVE n OIL LEVEL LO, OIL OVHT (above 200 C) and OIL PRESS LO (below 35 PSI); the DRIVE pb FAULT light (OVHT and PRESS LO); after 5 min of low level the generator switches off, after 5 min of low pressure or above 250 C the drive disconnects (then ELEC DRIVE n DISCONNECTED). Normal drive oil temperatures no longer reach 200 C at high N2.
- The **APU GEN A (B) pb-sw FAULT** light comes on when the GGPCU disconnects the APU generator.
- **SD ELEC AC and DC pages:** GEN, TR, STAT INV and RAT turn amber when they are failed.
- **Electric hydraulic pumps**: on the ground the green ones are inhibited only by engine 1 or 2 running, the yellow ones by engine 3 or 4; all are inhibited in flight (FCOM).
- Shared code: the engine generator drive oil model of the A32NX IDG (drive trait and oil failure types) is on this branch as the base of the A380X drive model; the A32NX IDG failures are not offered here (they are on core/failures-a32nx). The A32NX cabin pressure part is on core/failures-a32nx.

What you should see

Situation	What you should see
DRIVE 1 Oil Leak, engine 1 running	About 25 s after the leak starts the oil level is low (ELEC DRIVE 1 OIL LEVEL LO, shown only on the ground before the first engine start and after the last shutdown); about 50 s after, the pressure is low: ELEC DRIVE 1 OIL PRESS LO with the DRIVE 1 FAULT light and the SD ELEC AC page; 5 min later the drive disconnects (ELEC DRIVE 1 DISCONNECTED).
DRIVE 1 Oil Cooling Lost, engine 1 running	About 4 min later ELEC DRIVE 1 OIL OVHT with the DRIVE 1 FAULT light; about 100 s later (250 C) the drive disconnects.

Limitations and why

Limitation	Why	What to do
The leak and heating rates are design choices.	The FCOM gives no normal oil values.	-
A disconnected drive cannot be reconnected in flight.	FCOM DRIVE pb DISC.	Reconnect on the ground.

How to use

1. Drive oil alerts: follow the E/WD (GEN ... OFF or DRIVE ... DISC).
2. Manual cabin pressure: set MAN and turn the CABIN ALT TRGT or V/S TRGT selector to take control.

How to test

#	Do this	Expected result

1	A380X in cruise. flyPad > Failures > ATA 24 Electrical Power > **DRIVE 1 Oil Leak**. Wait 1 min.	ELEC DRIVE 1 OIL PRESS LO, DRIVE 1 FAULT light, SD ELEC AC page.
2	Wait 5 min more without action.	The drive disconnects: ELEC DRIVE 1 DISCONNECTED. Deactivate the failure.
3	Cruise. flyPad > Failures > ATA 24 Electrical Power > **DRIVE 2 Oil Cooling Lost**. Wait about 4 min.	ELEC DRIVE 2 OIL OVHT, DRIVE 2 FAULT light; about 100 s later the drive disconnects.
4	Cruise. flyPad > Failures > ATA 24 Electrical Power > **Static Inverter** or **TR 1**; SD ELEC DC page.	The failed item is amber. Deactivate it.
5	Cruise, pressurization: CABIN ALT MODE MAN.	The cabin altitude stays where it is until the CABIN ALT TRGT selector is turned; then it follows the selector.
6	On the ground, engines 3 and 4 running, 1 and 2 off. HYD panel: green electric pumps.	The green electric pumps can run; the yellow ones are inhibited.

What PR testers should check

- In MAN ALT / MAN V/S, the first selector turn jumps to the selector position.
- The drive oil failure timeline (level, pressure, disconnection).
- The APU GEN FAULT light during an APU start.

11. APU automatic shutdowns, smoke detection and cargo extinguishing alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 26 and ATA 49; APU panel; CARGO SMOKE panel; E/WD, STATUS | **Status:** Not sim-tested (unit tests: Rust ECB, smoke detection, cargo extinguishing and fire tests, failing with the logic disabled; SmokeAlerts.spec.ts (FWS side only), failure list specs).

What it is for

- **APU ECB automatic shutdowns** (FCOM PRO-ABN-ECAM-10-49): the APU shuts down by itself, with APU FAULT, on an ECB failure, an overspeed, a low oil pressure, a high oil temperature (A380X: on the ground only) and when its DC supply is lost above N 70 % (1 s confirmation). Four new flyPad failures: **APU electronic control box (ECB)**, **APU overspeed (automatic shutdown)**, **APU low oil pressure (automatic shutdown)**, **APU high oil temperature (automatic shutdown on ground)**.
- **Smoke detection** (FCOM DSC-26): the A380X FWS raises the SMOKE warnings and SMOKE ... DET FAULT cautions of 9 zones: FWD, AFT and BULK CARGO, L / R MAIN AVNCS, L / R UPPER AVNCS, AFT AVNCS and IFE BAY (for example SMOKE FWD CARGO SMOKE, SMOKE FWD CARGO DET FAULT), with their procedure lines.
- **Cargo fire bottles**: SMOKE FWD+AFT CARGO BTL 1 (2) FAULT when a bottle has a low pressure; STATUS INOP SYS FWD+AFT CRGs BTL 1 (2).
- New flyPad failures (ATA 26): **Forward / Aft / Bulk cargo smoke**, **Left / Right main avionics bay smoke**, **Left / Right upper avionics bay smoke**, **Aft avionics bay smoke**, **IFE bay smoke**, the smoke detectors 1 and 2 of each zone (for example **Forward cargo smoke detector 1**), **Cargo fire bottle 1 (2) low pressure**.

Limitations and why

Limitation	Why	What to do
The SMOKE alerts need `core/failures-a32nx`.	The A380X smoke detection model (shared fire model) and the CARGO SMOKE panel lights (SMOKE, DISCH, BTL, AGENT pbs) are on that branch; the alerts here read its L:vars.	Merge both; without it the smoke failures give nothing.

No smoke detection in the crew rests, lavatories, CWS, showers and facilities.	Those zones are not modelled.	-
No bottle diverter / control alerts (FWD (AFT) CARGO BOTTLES FAULT).	Not modelled.	-
Some APU shutdown causes are missing.	No model for them.	-

How to use

1. APU FAULT: set the APU MASTER SW OFF as the E/WD asks.
2. SMOKE warnings: follow the E/WD (ISOL VALVES, AGENT, CAB FANS ...), with core/failures-a32nx merged.

How to test

#	Do this	Expected result
1	A380X at the gate, APU running (APU MASTER SW ON, START). flyPad > Failures > ATA 49 Airborne Auxiliary Power > **APU overspeed (automatic shutdown)**.	The APU shuts down; APU FAULT with AUTO SHUTDOWN and -APU MASTER SW ... OFF; STATUS INOP SYS APU.
2	APU MASTER SW OFF.	The line is done; APU FAULT goes when the ECB is off. Deactivate the failure.
3	With core/failures-a32nx merged, at the gate: flyPad > Failures > ATA 26 Fire Protection > **Forward cargo smoke**.	Red SMOKE FWD CARGO SMOKE with its lines (ISOL VALVES, AGENT ... DISCH). Deactivate it.
4	At the gate: activate **Forward cargo smoke detector 1** and **Forward cargo smoke detector 2**.	SMOKE FWD CARGO DET FAULT (crew awareness). Deactivate both.
5	At the gate: activate **Cargo fire bottle 1 low pressure**.	SMOKE FWD+AFT CARGO BTL 1 FAULT; STATUS INOP SYS FWD+AFT CRGs BTL 1.

What PR testers should check

- No APU shutdown during the normal power transfers (EXT PWR, APU GEN, engine generators).
- The DISCH / AGENT / BTL light emissives (with core/failures-a32nx).
- The smoke failures from the flyPad.

12. bleed monitoring, leak detection, BLEED FAULT and LEAK alerts

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 36; overhead AIR panel (ENG BLEED pbs); SD BLEED; E/WD | **Status:** Not sim-tested (unit tests: A380 bleed_monitoring 12 Rust tests, failing with the logic disabled; BleedMonitoringAlerts, EngineBleedValveIndication and AbnormBleedConfig specs).

What it is for

- **Bleed monitoring** (FCOM DSC-36-10-20): the CPIOM-A closes an engine bleed valve on an overtemperature (260 C, or 245 C above 45 PSI, for 55 s; the HP valve too) or an overpressure (60 PSI, or 45 PSI above 245 C, for 15 s), held until the ENG BLEED pb is set OFF. A failed bleed opens the crossbleed valves at AUTO.
- **Leak detection** (DSC-36-10-60): a leak of an engine pylon duct (above 180 C) or of a wing duct (above 124 C) closes that bleed valve and the crossbleed valves of its side.
- The **ENG 1 (2)(3)(4) BLEED FAULT** light comes on for these cases and when a bleed valve is not in its commanded position (DSC-36-20).

- New alerts (PRO-ABN-ECAM-10-36): **AIR ENG 1 (2)(3)(4) BLEED FAULT**; **AIR ENG 1 (2)(3)(4) BLEED LEAK** and **AIR L (R) OUTR (INR) WING LEAK** with the AIR LEAK ISOLATED line. A BLEED FAULT also triggers AIR ABNORM BLEED CONFIG.
- New flyPad failures (ATA 36): **Engine 1 (2)(3)(4) bleed valve stuck closed**, **... stuck open**, **Engine 1 (2)(3)(4) pylon bleed duct leak**, **Left / Right outer / inner wing bleed duct leak**.
- Shared code: the bleed air failure types of the A32NX pneumatic failures come with it (prerequisite commit, no behaviour of its own). The A32NX BLEED page part is on core/failures-a32nx.

Limitations and why

Limitation	Why	What to do
No A380X pack alerts.	No pack valve or regulation failure model.	-
One leak loop per zone; no APU, pack bay or hot-air duct leak; no HP stage pressure low.	Not modelled.	-
A pylon leak is only detected at high power.	The duct must be pressurised enough to heat the loop.	Test it with high thrust.

How to use

1. Follow the E/WD: ENG BLEED pb OFF for a leak or a bleed fault; AIR LEAK ISOLATED tells when it is done.
2. Use the AIR ABNORM BLEED CONFIG lines (crossbleed, air flow, packs, wing anti-ice) after a bleed is lost.

How to test

#	Do this	Expected result
1	A380X in climb or cruise. flyPad > Failures > ATA 36 Pneumatic > **Left outer wing bleed duct leak**.	ENG 1 BLEED FAULT light; AIR L OUTR WING LEAK; the engine 1 bleed valve and the left and center crossbleed valves close (SD BLEED).
2	ENG 1 BLEED pb OFF, deactivate the failure.	The FAULT light goes off once the overheat is gone. ENG 1 BLEED pb back to AUTO.
3	Climb thrust. Activate **Engine 2 pylon bleed duct leak**.	ENG 2 BLEED FAULT light; AIR ENG 2 BLEED LEAK; bleed valve 2 and the left crossbleed valve close. Deactivate it, ENG 2 BLEED OFF then AUTO.
4	Activate **Engine 3 bleed valve stuck open**, then ENG 3 BLEED pb OFF.	ENG 3 BLEED FAULT light (the valve is not where it is commanded). Deactivate it, pb AUTO.

What PR testers should check

- No false AIR ENG BLEED FAULT on a fast thrust increase at take-off.
- The leak failures from the flyPad (the pylon leak needs high power to be detected).

13. landing gear and brakes

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 32; L/G lever, L/G GRVTY switch, A-SKID switch, AUTO BRK; SD WHEEL; E/WD | **Status:** Not sim-tested (unit tests: 8 Rust gear / brake tests, failing first, 4 EmergencyBrakeLaw tests, FwsBrakeAlerts.spec.ts cases).

What it is for

- The gear extension is inhibited above **280 kt** (FCOM DSC-32-10-20; was 260 kt, the A320 value).

- Hydraulic supply as in the FCOM: the nose gear on GREEN, each main gear assembly (body and wing) on GREEN and YELLOW: with GREEN or YELLOW lost the main gears no longer retract, and the gear is extended by gravity.
- The **gravity extension** takes about 70 s from the L/G GRVTY switch to the last gear downlocked (was about 50 s).
- **Emergency braking** (A-SKID off, or normal and alternate braking lost) follows the EBCU laws: 1 700 psi for taxi-out and take-off, 1 000 psi from lift-off until the PARK BRK handle is set ON (was 1 160 psi).
- The **RTO autobrake** no longer activates below 40 kt, but an active one continues to the stop.
- New alerts: **BRAKES A-SKID FAULT ON ALL L/G** (emergency braking with the A-SKID switch ON) and **BRAKES AUTO BRK FAULT** with the new flyPad failure **Brakes autobrake (AUTO BRK)**.

Limitations and why

Limitation	Why	What to do
The wing and body gears are still one gear per side.	The gear model has no separate wing gears.	-
With GREEN lost the body gear, with YELLOW lost the nose gear do not retract.	Known deviation of the shared LGCIU: the gears move only with all doors open.	-
No braking LEHGS, separate accumulators or BCS split; 16 BRAKES alerts still missing.	Not modelled.	-

How to use

1. Extend the gear below 280 kt (and below VLO).
2. For a gravity extension, set the L/G GRVTY switch DOWN and wait about 70 s.

How to test

#	Do this	Expected result
1	A380X at the gate, engines running, A-SKID ON. flyPad > Failures > ATA 32 Landing Gear > **Brakes normal braking (BCS NORM BRK)** and **Brakes alternate braking (BCS ALTN BRK)**.	Emergency braking: BRAKES A-SKID FAULT ON ALL L/G (with the alerts of the two failures). Deactivate both.
2	At the gate: activate **Brakes autobrake (AUTO BRK)**.	BRAKES AUTO BRK FAULT; SD WHEEL page. Deactivate it.
3	In flight, gear up, below 250 kt: L/G GRVTY switch DOWN, start a timer.	The last gear is downlocked after about 70 s; SD WHEEL L/G GRVTY EXTN IN PROGRESS meanwhile.
4	Take-off roll with RTO armed, reject below 40 kt.	The RTO autobrake does not activate.
5	After landing with A-SKID OFF, brake with the pedals before setting the parking brake.	Emergency braking limited to 1 000 psi.

What PR testers should check

- The gravity extension timing in flight.
- The L/G alerts and indications with GREEN or YELLOW lost.
- The landing law after touchdown with A-SKID OFF.

14. HP fuel valve, FADEC faults, motoring and auto-idle

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 73 and ATA 79; ENG MASTER, ENG START; thrust reversers; E/WD, SD ENGINE | **Status:** Not sim-tested (unit tests: Rust systems / a380 tests, 10 failing first, and module tests; EngineMonitoringAlerts.spec.ts; failure list spec).

What it is for

- New flyPad failures **Engine 1 (2)(3)(4) HP fuel valve stuck** (ATA 73): the valve stays where it is; stuck closed, the engine gets no fuel and does not light up. When the valve position disagrees with the FADEC command for 2 s: **ENG 1 (2)(3)(4) HP FUEL VLV FAULT** (FCOM PRO-ABN-ECAM-10-70; master caution, ENGINE page) and the ENG FAULT light.
- New FADEC fault failures with their crew awareness alerts (no aural, no master light): **Engine n fuel strainer clog** - ENG n FUEL STRAINER CLOGGED; **Engine n fuel filter monitoring** - ENG n FUEL FILTER MONITORING FAULT; **Engine n oil chip (metal particles)** (ATA 79) - ENG n OIL CHIP DETECTED; **Engine n FADEC minor fault (internal sensor)** - ENG n MINOR FAULT; **Engine n overthrust protection** - ENG n OVTHR PROT LOST; **Engine n pressure and temperature probes** - ENG n SENSOR FAULT. 28 A380X failures in all.
- **Auto-idle**: engines 1 and 4 are held at idle while a thrust reverser operates (FCOM).
- Pre-start motoring of a hot engine (shared engine start code, as on core/failures-a32nx).

What you should see

Situation	What you should see
FUEL STRAINER CLOGGED	Shown in flight phases 1, 2, 11, 12 (inhibited 3 to 10).
FUEL FILTER MONITORING FAULT, OIL CHIP DETECTED	Shown only in phases 11 and 12 (after landing).
MINOR FAULT, OVTHR PROT LOST, SENSOR FAULT	Shown in phases 1, 11 and 12 (inhibited 2 to 10).
HP FUEL VLV FAULT	Inhibited in phases 3 to 7, 9 and 10.

Limitations and why

Limitation	Why	What to do
Other ENG alerts are still missing.	They need engine models the sim does not have.	-
No approach idle; no A380X derates.	No FCOM value; no A/THR model for the derates.	-

How to use

1. HP FUEL VLV FAULT during a start: ENG MASTER OFF (and ENG MAN START OFF for a manual start), as the E/WD asks.
2. The crew awareness FADEC alerts need no action in flight; report them after the flight.

How to test

#	Do this	Expected result
1	A380X at the gate, engines off. flyPad > Failures > ATA 73 Engine - Fuel And Control > **Engine 1 HP fuel valve stuck**. Start engine 1.	No fuel flow, no light-up; after 2 s ENG 1 HP FUEL VLV FAULT and the ENG 1 FAULT light. ENG MASTER 1 OFF, deactivate the failure.

2	At the gate, aircraft powered, engines off (phase 1). Activate **Engine 2 fuel strainer clog**, **Engine 3 FADEC minor fault (internal sensor)** and **Engine 4 pressure and temperature probes**.	ENG 2 FUEL STRAINER CLOGGED, ENG 3 MINOR FAULT, ENG 4 SENSOR FAULT (crew awareness, no chime). Deactivate them.
3	After landing, reversers deployed.	Engines 1 and 4 stay at idle while engines 2 and 3 give reverse thrust.
4	Engine running: activate **Engine 1 HP fuel valve stuck**, then ENG MASTER 1 OFF.	The engine still stops (the LP valve closes); ENG 1 HP FUEL VLV FAULT.

What PR testers should check

- Hot restart motoring.
- An A380X HP valve stuck open still stops the engine at ENG MASTER OFF.
- The outboard auto-idle in reverse.

15. LP valve, emergency outer tank transfer and fuel leak detection

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 28; overhead FUEL panel (EMER OUTR TK XFR); SD FUEL; E/WD | **Status:** Not sim-tested (unit tests: Rust gravity_feed 16 and engine_lp_valve 5, 8 failing first; FwsFuelLeakLpValveAlerts, FuelLeak, FuelEmerOuterTransfer and FuelPumpsAndValves specs).

What it is for

- New flyPad failures **ENG 1 (2)(3)(4) LP fuel valve jammed in place**: **FUEL ENG n LP VLV FAULT** (FCOM PRO-ABN-ECAM-10-28) when the valve stays closed while commanded open (engine start) or stays open while commanded closed (engine shutdown), on the ground; before take-off -ENG n MASTER ... OFF. Jammed closed, the engine cannot start. STATUS INOP SYS ENG n LP VLV.
- The **EMER OUTR TK XFR** pb-sw opens the emergency outer tank transfer valves: the outer tank fuel flows by gravity into feed tanks 1 and 4 (flight_model.cfg gets the missing Feed 4 input line).
- **FQMS fuel leak detection** (DSC-28-40): slow leak (block fuel minus (FOB + fuel used) above 15 450 lb) and fast leak (FOB decrease minus fuel used above 4 400 lb in 30 min); **FUEL LEAK DETECTED** (phases 2 and 8) with the full FUEL LEAK procedure; **FUEL LEAK DET FAULT** (phase 8) with STATUS INOP SYS FUEL LEAK DET. New flyPad failures **Feed tank 1 (2)(3)(4) leak**.
- Shared code: the LP valve jam (engine_lp_valve.rs) is the same as on core/failures-a32nx, where the A32NX part is.

Limitations and why

Limitation	Why	What to do
A full MSFS restart is needed.	flight_model.cfg changes.	Restart the sim after installing.
Other fuel alerts are still missing.	They need temperature, overflow, inerting or FQMS transfer models.	-

How to use

1. FUEL LEAK DETECTED: follow the FUEL LEAK procedure on the E/WD.
2. EMER OUTR TK XFR pb ON to move the outer tank fuel into feed tanks 1 and 4.

How to test

#	Do this	Expected result

1	A380X at the gate, engines off. flyPad > Failures > ATA 28 Fuel > **ENG 1 LP fuel valve jammed in place**. ENG MASTER 1 ON (start).	The engine does not start; FUEL ENG 1 LP VLV FAULT with -ENG 1 MASTER ... OFF. MASTER OFF, deactivate the failure.
2	Cruise with fuel in the outer tanks. FUEL panel: EMER OUTR TK XFR pb ON. SD FUEL page.	The outer tanks empty into feed tanks 1 and 4. pb OFF.
3	Cruise. Activate **Feed tank 1 leak**; wait up to 30 min.	FUEL LEAK DETECTED once the lost fuel is 4 400 lb more than the fuel used; follow the FUEL LEAK procedure. Deactivate the failure.
4	Normal flight without failure.	No FUEL LEAK DETECTED.

What PR testers should check

- EMER OUTR TK XFR into feed tanks 1 and 4.
- No false FUEL LEAK DETECTED.

16. anti-ice valves, wing anti-ice, oxygen, anti-ice key events

> **Aircraft:** A380X and A32NX (engine anti-ice valves, key events) | **Where:** flyPad > Failures > ATA 30 and ATA 35; overhead ANTI ICE panel; SD DOOR/OXY; E/WD | **Status:** Not sim-tested (unit tests: Rust engine_anti_ice, A380 ice_rain_protection and oxygen tests, failing first; AntiIceValveAlerts, OxygenAlerts and FadecWingAntiIce specs; the wasm key event mapping tests compile but cannot run on the host).

What it is for

- **A380X wing anti-ice system** (FCOM DSC-30): the WING ANTI ICE pb commands two valves per wing on bleed air: 30 s ground test, then 10 min inhibition; inhibited during the take-off until the thrust reduction; FAULT light. The flight controls and the FADECs now get the real wing anti-ice state.
- **Engine anti-ice valves** (both aircraft, shared model): open with the pb ON, or unpowered once the HP air is above 10 psig; 2 s travel; FAULT light. The MSFS engine anti-ice follows the valves.
- New A380X alerts (PRO-ABN-ECAM-10-30): **A-ICE ENG 1 (2)(3)(4) VLV CLOSED / VLV OPEN**, **A-ICE L (R) WING VLV CLOSED / VLV OPEN / VLV PRESS HI / VLV PRESS LO**.
- **A380X oxygen** (PRO-ABN-ECAM-10-35): cockpit and cabin bottle pressures on the DOOR/OXY page; **OXY CKPT BOTTLE PRESS LO**, **OXY CABIN BOTTLE PRESS LO**, **OXY CABIN VLV CLOSED**.
- New flyPad failures: ATA 30 **Left (Right) wing anti-ice valves stuck closed**, **Left (Right) wing outer anti-ice valve stuck open**, **Engine 1 (2)(3)(4) anti-ice valve stuck closed / stuck open**; ATA 35 **Cockpit oxygen bottle leak**, **Cabin oxygen bottles leak**, **Cabin oxygen regulator supply valve stuck closed**.
- **Key events** (both aircraft): MSFS anti-ice events (ANTI_ICE_TOGGLE_ENGn, ANTI_ICE_SET_ENGn, ANTI_ICE_ON / OFF / SET / TOGGLE, TOGGLE_STRUCTURAL_DEICE) now move the ENG and WING ANTI ICE pbs, so key bindings and hardware panels keep working.
- The A32NX engine anti-ice valves (with their ENG ANTI ICE pb FAULT light) come with the shared key event code: without them the A32NX would not build. The A32NX probe and window heat, alerts and failures are on core/failures-a32nx.

Limitations and why

Limitation	Why	What to do
No A380X probe / window heat model and its 15 alerts; no wing anti-ice on the SD BLEED page.	Not done yet.	-
No ADR probe heat bits.	Not done yet.	-

> **NOTE**
> Fix-up commit de9a236e6 removes a native stub of the systems WASM that these changes left unused (the workspace clippy check failed on it). No effect for the user.

How to use

1. Use the ENG and WING ANTI ICE pbs (or your key bindings / hardware panel) as usual.
2. On an A-ICE or OXY alert, follow the E/WD.

How to test

#	Do this	Expected result
1	A380X on the ground, engines running. WING ANTI ICE pb ON.	The wing valves open for the 30 s ground test, then close; ON again within 10 min: no new test.
2	Cruise. flyPad > Failures > ATA 30 Ice And Rain Protection > **Engine 1 anti-ice valve stuck closed**; ENG 1 ANTI ICE pb ON.	ENG 1 ANTI ICE FAULT light; A-ICE ENG 1 VLV CLOSED. pb OFF, deactivate the failure.
3	Cruise. Activate **Left wing anti-ice valves stuck closed**; WING ANTI ICE pb ON.	A-ICE L WING VLV CLOSED. pb OFF, deactivate it.
4	flyPad > Failures > ATA 35 Oxygen > **Cockpit oxygen bottle leak**; SD DOOR/OXY page.	The cockpit bottle pressure drops; then OXY CKPT BOTTLE PRESS LO. Deactivate it.
5	Bind a key to the MSFS "toggle engine 1 anti-ice" event and press it (either aircraft).	The ENG 1 ANTI ICE pb moves; no toggle loop.

What PR testers should check

- Key binding / hardware panel toggles of ENG and WING anti-ice: no loop, no toggle storm at flight load.
- A32NX ENG A.ICE FAULT light before about 25 % N2.
- A380X wing pb ground test and take-off inhibition.
- The oxygen leak failure on the DOOR/OXY page.

17. avionics network failures, SFCC channels, TAIL STRIKE

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 27, ATA 31 and ATA 42; E/WD, SD, STATUS | **Status:** Not sim-tested (unit tests: Rust avionics network and SFCC channel tests, 8 failing first; FwsTailStrikeAlerts.spec.ts and failure list specs, 14 failing first).

What it is for

- New ATA 42 failures (FCOM DSC-42-10/20) of the 16 **AFDX switches** (for example **AFDX switch 1 (network A)**), the 8 **IOMs** (**IOM A1** to **IOM A8**) and the 22 **CPIOMs** (for example **CPIOM C1 (FWS, FCDC)**, **CPIOM F1 (fuel)**). A failed switch is rerouted around; a failed module is no longer available, and the EWD, SD, PFD and FWS parts that already watch the CPIOM availability react to it.
- New ATA 27 failures **SFCC 1 (2) flap channel** and **SFCC 1 (2) slat channel**: the failed channel stops its motor (flaps or slats at half speed); F/CTL FLAP (SLAT) SYS 1 (2)(1+2) FAULT now have a trigger.
- **TAIL STRIKE** caution (PRO-ABN-ECAM-10-27-10, LAND ANSA) after a detected contact, and **TAIL STRIKE DET FAULT** (STATUS INOP SYS TAIL STRIKE DET) with the new failure **Tail strike detector** (ATA 31). Contact = pitch 11.5 deg or more with a main gear compressed (the FCOM LIM-10 geometry limit, design choice); it is remembered until the next flight (phase 1).

Limitations and why

Limitation	Why	What to do

Network losses have no effect on FMS, datalink, braking, steering, transponder or PFD yet.	Not modelled.	-
No actuator, sidestick, THS or WTB failures; no backup control (BCM / BPS); no ALT 1C reconfiguration.	They need the flight control model (Simulink).	-
The tail strike contact is a pitch / gear rule.	The sim has no fuselage contact sensor.	-

How to use

1. Activate the failures in the flyPad ATA 27, 31 and 42 lists.
2. After a TAIL STRIKE, land as soon as possible (LAND ANSA).

How to test

#	Do this	Expected result
1	A380X in flight below VFE. flyPad > Failures > ATA 27 Flight Controls > **SFCC 1 flap channel**; select the next flap position.	F/CTL FLAP SYS 1 FAULT; the flaps move at half speed. Deactivate it.
2	At the gate. flyPad > Failures > ATA 31 Indicating / Recording System > **Tail strike detector**.	TAIL STRIKE DET FAULT; STATUS INOP SYS TAIL STRIKE DET. Deactivate it.
3	Normal take-off and landing.	No TAIL STRIKE.
4	Cruise. flyPad > Failures > ATA 42 Integrated Modular Avionics > **AFDX switch 1 (network A)**.	Nothing lost (rerouted).
5	Activate **CPIOM C1 (FWS, FCDC)** and **CPIOM C2 (FWS, FCDC)**.	Note what the FWS, E/WD and SD lose and report it. Deactivate both.

What PR testers should check

- No false TAIL STRIKE on normal rotations and flares.
- Half-speed flaps / slats with a channel failure.
- CPIOM C1 / C2 or AFDX switch failures and their FWS / EWD / SD loss.

18. reactive windshear, GPWS mode 2, alpha floor Mach limit

> **Aircraft:** A380X | **Where:** PFD; GPWS aurals | **Status:** Not sim-tested (unit tests: Rust egpws 3, C++ alpha_floor_mach_inhibit_test, vitest windshear 13 (ReactiveWindshearDetection.spec.ts), GpwsMode2 8; all failing without the change).

What it is for

- **Reactive windshear** (FCOM DSC-22): the PRIMs detect a decreasing performance windshear at take-off and landing below 1 300 ft, CONF 1 or more: red WINDSHEAR on both PFDs (L:A32NX_REACTIVE_WINDSHEAR_WARNING). The FCOM gives no formula: F-factor above 0.13, filtered (design choice). The voice comes with the voices feature below.
- **Alpha floor** is inhibited above M 0.5 (was M 0.6), in the generated PRIM code.
- **GPWS mode 2** (excessive terrain closure rate) works again with a filtered radio altitude rate: TERRAIN then PULL UP on the PFD with the PULL UP aural, TERRAIN only in landing configuration. The PFD now shows the PULL UP of

mode 1 (excessive rate of descent): the mode 2 bit overwrote it.

- Shared code: the windshear detector, the PFD warning and the egpws Rust are the same as on core/failures-a32nx, where the A32NX part (FAC windshear, GPWS test, TCAS failure) is.

Limitations and why

Limitation	Why	What to do
No REAC W/S DET FAULT alert, SRS escape guidance, FLS modes, THR DES / DCLB, TAD / TCF or predictive windshear.	They need models, terrain or wind data.	-
The Simulink PRIM model still has M 0.6.	Only the generated code is changed.	Change the model before a regeneration.

How to use

1. On WINDSHEAR or PULL UP, apply the FCOM escape manoeuvre.

How to test

#	Do this	Expected result
1	A380X approach in gusty weather, CONF 1 or more, below 1 300 ft; normal turns.	No WINDSHEAR without a real shear.
2	Low flight over rising terrain, gear up, high terrain closure rate.	TERRAIN then PULL UP on the PFD with the PULL UP aural.
3	High rate of descent close to the ground (mode 1).	PULL UP on the PFD.
4	Cruise above M 0.5 at a high angle of attack (for example a tight high-altitude turn).	No alpha floor activation.

What PR testers should check

- No false WINDSHEAR in gusts or turns.
- A380X mode 2 over rough terrain.

19. NAV FM / GPS POS DISAGREE and NAV LS TUNING DISAGREE

> **Aircraft:** A380X | **Where:** E/WD, STATUS | **Status:** Not sim-tested (unit tests: FwsNavPositionAlerts.spec.ts, failing first).

What it is for

- **NAV FM / GPS POS DISAGREE** (FCOM PRO-ABN-ECAM-10-34-10) now triggers when the FMS finds its position too far from the GPS position (latitude-dependent FCOM thresholds; L:A32NX_FM_GPS_POS_DISAGREE of the FM position model of `core/gps`, merged here through core/a380x-elec): master caution, -ACFT POS ... CHECK, STATUS INFO FLS : RAW ONLY. Inhibited in phases 1, 3 to 6 and 12.
- **NAV LS TUNING DISAGREE** when the two MMRs (sim NAV 3 and NAV 4) disagree on frequency or course: master caution, phases 8 and 9 only; STATUS INOP SYS CAT 2, GLS AUTOLAND.

Limitations and why

Limitation	Why	What to do

The approach mode is not compared for LS TUNING DISAGREE.	The sim receivers have no approach mode.	-

How to use

1. On NAV FM / GPS POS DISAGREE, check the aircraft position (POSITION MONITOR page) as the E/WD asks.

How to test

#	Do this	Expected result
1	A380X normal flight with GPS.	No NAV FM / GPS POS DISAGREE.
2	ILS approach tuned by the FMS, below 1 500 ft.	No NAV LS TUNING DISAGREE.
3	If NAV FM / GPS POS DISAGREE shows, open STATUS.	INFO FLS : RAW ONLY.

What PR testers should check

- No false NAV FM / GPS POS DISAGREE or LS TUNING DISAGREE in normal flights.

20. A380X FWS voices: sidestick priority, intermediate RA callouts, WINDSHEAR, TERRAIN, GLIDE SLOPE, PITCH PITCH, COMPANY MESSAGE

> **Aircraft:** A380X | **Where:** Cockpit aurals; sidestick pbs | **Status:** Not sim-tested (unit tests: SidestickPriorityAurals, IntermediateHeightCallout, FwsVoiceCallouts, AutoCallOuts, FwsSidestickPriority, FwsWindshearAural, FwsVoiceSounds, LegacyGpws specs (27 failing first); FwsPitchAural 7, FwsCompanyMessageAural 5 and more AutoCallOuts / LegacyGpws / FwsVoiceSounds cases (24 failing first)).

What it is for

- **PRIORITY LEFT / PRIORITY RIGHT** (FCOM DSC-27-10-20) when a pilot holds the sidestick pb with the priority for 1 s (a short press to disconnect the autopilot does not call it; it waits during a GPWS alert).
- **Intermediate RA callout**: below 400 ft, the height rounded to 10 ft (for example TWO HUNDRED SEVENTY) 11 s after the previous callout; below 100 ft, the height to the foot (for example FORTY SEVEN) 4 s after it.
- **WINDSHEAR** said three times with the reactive windshear warning; it also silences the GPWS alerts.
- **TERRAIN** for GPWS mode 2 and **GLIDE SLOPE** (louder when hard) for mode 5.
- **PITCH PITCH** (DSC-22-27-20) in a manual landing below 400 ft RA with the pitch above 12 deg, once per event (the FWS computes it: the PRIM output is always false).
- **COMPANY MESSAGE** (DSC-46-40-20) when a company (AOC) message arrives, except in take-off and landing and below 10 000 ft in phase 8.
- The clips are made with tools/fws-voice (Piper TTS, public-domain LJ Speech voice) and play from the Wave Data slots of the A380X sound.xml.

Limitations and why

Limitation	Why	What to do
TERRAIN and GLIDE SLOPE may stay silent.	The A380X Wwise sound bank lacks 13 A32NX events (upstream large-files branch align-a380x-sounds).	Wait for that upstream merge.
No DUAL INPUT voice.	The F/O sidestick sends no input in the sim.	-

No SPEED SPEED, SELCAL or time marker voices.	No FCOM threshold / PRIM output, no SELCAL source, the FCOM contradicts itself on the time marker.	-
No COMPANY ALERT or CALL COMPANY.	The AOC has no urgent or call-company messages.	-
A full MSFS restart is needed.	sound.xml changes.	Restart the sim after installing.

How to use

1. Nothing to set: the voices come by themselves.

How to test

#	Do this	Expected result
1	A380X in flight, AP ON. Hold the captain sidestick pb 2 s.	AP disconnects; PRIORITY LEFT is said once. A short press only disconnects (no voice).
2	Manual approach: from 400 ft RA down to the touchdown.	The usual callouts plus the intermediate heights (rounded to 10 ft above 100 ft, to the foot below).
3	Manual landing: pitch above 12 deg below 400 ft RA (for example a hard flare).	PITCH PITCH, once.
4	In cruise, with Hoppie ACARS set up: receive a company telex.	COMPANY MESSAGE (with the COMPANY MSG memo).
5	GPWS mode 2 or mode 5 situation.	TERRAIN or GLIDE SLOPE (if the A380X sound bank has the events).

What PR testers should check

- The volume of the WAV clips against the Wwise voices, and their retrigger.
- The A380X Wave Data slots play at all (warning, misc and instrument slots).
- The bank + Piper voice mix below 100 ft (gap, volume); the intermediate callouts are not too chatty.
- PITCH PITCH once per manual flare; COMPANY MESSAGE per Hoppie telex.
- The TERRAIN / GLIDE SLOPE events are present in the A380X bank.

21. No BRAKES BTV FAULT while an inner engine is not running

> **Aircraft:** A380X | **Where:** E/WD; STATUS | **Status:** Not sim-tested (unit tests: FwsBrakeAlerts.spec.ts, 6 new, failing first).

What it is for

The FCDC sets BTV LOST when engine 2 or 3 is not running, and the FWS raised BRAKES BTV FAULT (single chime, MASTER CAUT) straight from it, for example during the engine start on the ground. The FCOM trigger is "The BTV auto-brake mode is failed" (PRO-ABN-ECAM-10-32); an inner engine out makes BTV inoperative through the ENG SHUT DOWN STATUS. The caution now needs both inner engines running; STATUS INOP SYS BTV and LDG DIST AFFECTED still follow the FCDC BTV LOST.

How to use

1. Nothing to set.

How to test

#	Do this	Expected result

1	A380X cold and dark at the gate: start the engines one by one.	No BRAKES BTV FAULT and no single chime from it at any time.
2	In flight: shut down engine 2.	ENG 2 SHUT DOWN; STATUS INOP SYS BTV; no BRAKES BTV FAULT. Relight.

22. Engine exceedance marks only on displayed values

> **Aircraft:** A380X | **Where:** E/WD EGT and N1 gauges; SD ENGINE vibrations | **Status:** Not sim-tested (unit tests: EgtLimits, N1 and EngineVibration specs, failing first).

What it is for

The red N1 / EGT exceedance marks and the N3 red cross latched any value, also while the gauge showed XX (FADEC not sending) or on an infinite value, and a NaN after an exceedance put the mark at NaN. The FCOM shows XX for a value that is not available. The exceedance memories now take only displayed, finite values (E/WD gauge active flag, SD FADEC powered) and keep the highest value through a NaN. The A32NX part is on core/failures-a32nx.

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A380X cold and dark, then power up and start the engines.	No red exceedance mark or red cross on the E/WD N1 / EGT gauges or the SD ENGINE page.
2	Load a flight in cruise.	No red mark.

23. XFR pushbuttons start in AUTO and drive the manual fuel transfers

> **Aircraft:** A380X | **Where:** Overhead FUEL panel (OUTR, MID, INR TK XFR, TRIM TK XFR pbs); SD FUEL | **Status:** Not sim-tested (unit tests: FuelManualTransfer.spec.ts 27, with all AUTO commands and lineups checked against flight_model.cfg / systems.cfg, and FuelEmerOuterTransfer gravity cases, failing first).

What it is for

The OUTR, MID and INR TK XFR and TRIM TK XFR pb L:vars were never set at load (read 0 = MAN) and MAN did nothing. Now (FCOM DSC-28-40, "fuel transfers are fully automatic"):

- They start in AUTO at every flight load (released, no light), also with the hangar, taxi and climb flights.
- MAN (FWD for the trim tank) suspends the automatic transfers and lines up the FCOM manual transfer: OUTR outer tanks into the feed tanks (with an outer tank pump OFF, the emergency valves feed tanks 1 and 4 by gravity); INR and MID with their aft pumps into the feed tanks; TRIM FWD trim tank into the inner tanks.
- Back to AUTO restarts the automatic logic. Jettison and refuel suspend the manual transfers; no lineup takes fuel out of a feed tank.

Limitations and why

Limitation	Why	What to do
No AUTO GND XFR.	The FCOM rule is incomplete and flight_model.cfg has no wing-to-trim line.	-

No outer gravity transfer to feed tanks 2 and 3, no XFR FAULT lights, no MAN XFR ECAM procedures.	Not done.	-
The jettison suspension reads L:A380X_FUEL_JETTISON_IN_PROGRESS.	core/fuel (its jettison signals) is not on this branch.	With core/fuel merged it uses them.

How to use

1. Leave the XFR pbs in AUTO for normal operation.
2. Press one to MAN (FWD) for a manual transfer, as the FCOM asks; press it again for AUTO.

How to test

Test this first, before the other fuel tests.

#	Do this	Expected result
1	Load the A380X cold and dark, then at the apron, then in flight. Overhead FUEL panel.	All four XFR pbs released, no light; on the ground no tank level changes by itself.
2	In flight: OUTR TK XFR pb MAN; SD FUEL page.	The outer tanks flow into the feed tanks.
3	Back to AUTO; then INR TK XFR MAN.	The automatic logic restarts; with INR MAN the inner tanks flow into the feed tanks (the right inner pump runs for the right side).
4	MID TK XFR MAN, then TRIM TK XFR FWD.	Mid tanks into the feed tanks; trim tank into the inner tanks. All back to AUTO.

What PR testers should check

- First: at load (cold and dark, apron, in flight) all four XFR pbs released / no light and no tank level change on the ground.
- The flow of each pb on the SD FUEL page, and back to AUTO.
- INR MAN runs the right inner pump (pump index swap).